

I evaluated two models: Qwen2.5-VL-3B-Instruct (Qwen Team, 2025) and ChatGPT 5.4 (OpenAI, 2026), with the ScanNet (Dai et al., 2017) dataset, and looked at two specific VSTI-Bench (Bitton et al., 2023) tasks: `camera_movement_direction` and `camera_obj_abs_dist`.

Table 1: Aggregate Model Performance Comparison

Model	Task Category	Metric	Score
Qwen2.5-VL	Camera Movement	ACC	31.9%
	Absolute Distance	MRA	0.68
ChatGPT 5.4	Camera Movement	ACC	36.6%
	Absolute Distance	MRA	0.63

Now an example failure: Here the ground truth is 1.1, but Qwen2.5-VL estimated it to be 1.8 meters and ChatGPT

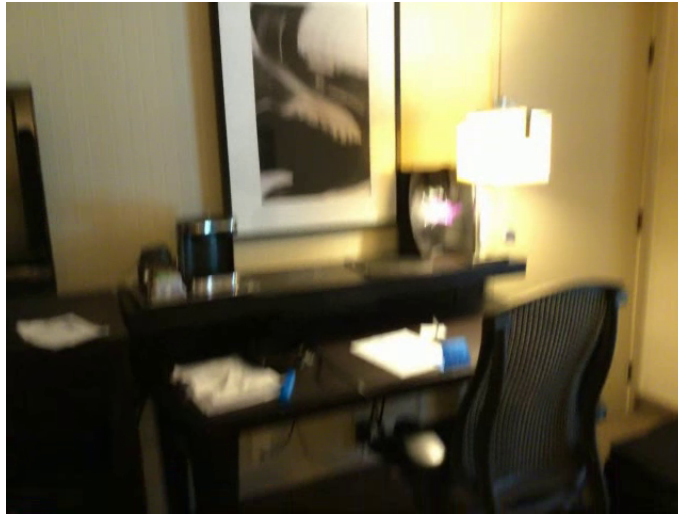


Figure 1: VSTI-Bench Task 1975

5.4 estimated it to be 2.0 meters.

The primary failure modes expose a fundamental architectural limitation: VLMs struggle to infer rigid 3D geometry purely from 2D pixel patches.

References

- A. Dai et al., “ScanNet: Richly-annotated 3D Reconstructions of Indoor Scenes,” *Proc. CVPR*, 2017.
- Qwen Team, “Qwen2.5-VL Technical Report,” *arXiv:2502.13923*, 2025.
- OpenAI, “ChatGPT (Version 5.4),” *OpenAI Blog*, 2026.
- Y. Bitton et al., “VisIT-Bench: A Benchmark for Vision-Language Instruction Following,” *Proc. NeurIPS*, 2023.